ch: 7 Hypothesis Testing : (اختبار الفرضيات)

Def : (1.) A statisical Hypothesis : فرضية احصائية
is an assumption about apopulation Paramer.(s)

This assumption may true or false. μ σ^2

(2.) Hypothesis testing : procedures to accept قبول or reject رفض
statisical Hypothesis.

There Are two types of statistical Hypothesis.

(A.) Null Hypothesies : (H_0) الفرضية البديهية
statement of Zero or No change عدم التغير
(الفرضية الصفرية (العدم

* If the original claim includes inequality ($\leq$ $\geq$ $=$)
it's is the null hypothesis.

(The decision is basedon the null hypothesis)

(B.) Alterative Hypothesis : النظرية البديلة

B.) Alterative Hypothesis : النظرية البديلة

(H_a or H_1)

Statement wich is true If the null hypothesis if false

Test statistic on the data.

الاختبار الاحصائي

accept H_0

reject H_0 ⇓ accept H_1

Ex: Compar two Population Means μ_1 & μ_2

$H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 \neq \mu_2$

* Types of Errors in hypothesis Testing : أنواع الأخطاء

		Truth (for population studied)	
		H_0 is true	H_0 is false
Decision (Based on sample)	reject H_0	Type I error α	correct degion
	accept H_0 (fail to reject)	correct decision	Type II error β

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* $\alpha = P(\text{getting type I error})$

$\alpha = (\text{reject } H_0 \mid H_0 \text{ is true})$

* أنت رفضت الـ H_0 على الرغم من أن H_0 صحيحة

** $\beta = P(\text{getting type II error})$

$= P(\text{accept } H_0 \mid H_0 \text{ is false})$

أنت قبلت الـ H_0 على الرغم من H_0 خاطئة

** Power of astarisical test $= 1 - \beta =$

(قوة الاختبار الاحصائي) $= P(\text{reject } H_0 \mid H_0 \text{ is false})$

Ex: Let $X_1, X_2, \ldots, X_n$ denote arandom sample of size $n = 25$ from anormal Population with varince $\sigma^2 = 4$, whose mean is known to either 0 or 1.

Test: $H_0 : \mu = 0$

$H_1 : \mu = 1$

we reject H_0 if $\bar{x} > 0.4$

1) Find the prob of getting type I error (α)? $\boxed{56}$
2) " " " " " " II " (β)?

Sol $\boxed{1}$ $\alpha = P(\text{reject } H_0 \mid H_0 \text{ is true})$

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$$= P(\bar{x} > 0.4 \mid \mu = 0)$$

$$= 1 - P(\bar{x} < 0.4 \mid \mu = 0)$$

$$= 1 - P\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < \frac{0.4 - 0}{2/5}\right) = 1 - P(Z < 1) = 1 - .8413$$

$$= .1587$$

$\boxed{2}$ $\beta = P(\text{getting type II error})$

$$= P(\text{accept } H_0 \mid H_0 \text{ is false})$$

$$= P(\bar{x} < 0.4 \mid \mu = 1)$$

$$= P\left(\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}}\right) < \frac{0.4 - 1}{2/5}\right) = P(Z < -1.5) = .0668.$$

H.W we reject H_0 if $\bar{x} > 0.5$

The relation between α & β As follows

As $\alpha \downarrow \Rightarrow \beta \uparrow$

As $\beta \downarrow \Rightarrow \alpha \uparrow$

(cannot minimize both types of error)

الجواب :

$\bar{x} > 0.4 \Rightarrow \alpha = 0.1587$, $\beta = .0668$

$\bar{x} > .5 \Rightarrow \alpha = .1056$, $\beta = .1056$

<u>Def</u> Level of significance $\Rightarrow \alpha$

مستوى الدلالة

the maximum prob that we have a type I error

$P(\text{reject } H_0 \mid H_0 \text{ is true})$

Usually $\boxed{\begin{array}{l}\alpha = 0.01 \\ \alpha = 0.05 \\ \alpha = 0.1\end{array}}$

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Hypothesis Testing of a Population Mean (μ)

Method 1: The Rejection Region Method
(using Z-statistic or t-statistic)

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Method 2: The P-value Method.
⇓ الشرح

Method :- Remark: let $x_1, x_2, \ldots, x_n$ be a random sample from normal population with mean μ & variance σ^2

$$X : N(\mu, \sigma^2)$$

$H_0 : \mu = \mu_0$ (known) against $H_1 : \mu \neq \mu_0 \rightarrow$ two-tailed
$\mu > \mu_0 \rightarrow$ right-tailed
$\mu < \mu_0 \rightarrow$ left-tailed

Steps to perform a test of Hypothesis Using Test-statistic
خطوات اكمل : اختبار الاحصاء

step (1): state the null hypotheses (H_0) & the alternative hypotheses (H_1)

step (2): Determine the rejection (رفض) & non-rejection (قبول) (accept) regions & calculate Z-critical or t-critical

Step (3): calculate the test statistic from the formula

توزيع طبيعي

$$\rightarrow Z_{statistic} = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \text{ or } Z_{statistic} = \frac{\overline{X} - \mu_0}{S/\sqrt{n}}, \; n \text{ is large}, \; n \geq 30$$

اللي بحدد ال n

$$\rightarrow t_{statistic} = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}, \; n \text{ is small}, \; n < 30$$

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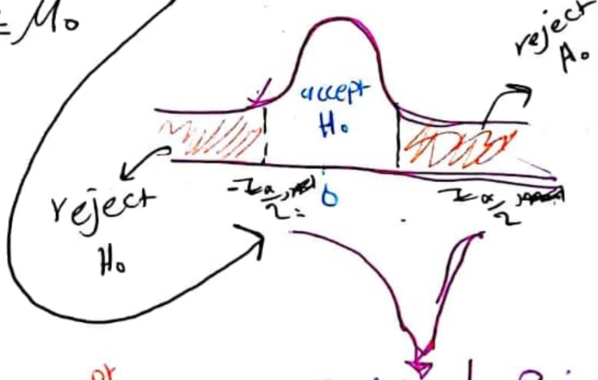
Step (4): Make the decision:-

A. Z-statistic; Two-Tailed → بقسمه على 2

$H_0 : \mu = \mu_0$

$H_1 : \mu \neq \mu_0$



Critical Points

نقاط حرجة

reject H_0 if

$$Z > Z_{\frac{\alpha}{2}} \text{ or } Z < -Z_{\frac{\alpha}{2}}$$

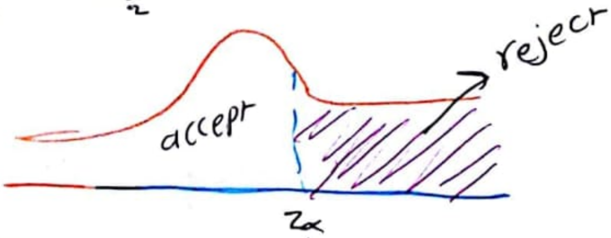
$$\rightarrow |Z| > Z_{\frac{\alpha}{2}}$$

تعريف

Right-Tailed

$H_0 : \mu = \mu_0$

$H_1 : \mu > \mu_0$



reject ~~the~~ H_0 if $Z > Z_\alpha$

بدون ال قسمة على 2

left - Tailed

$H_0 : \mu = \mu_0$

$H_1 : \mu < \mu_0$

$-Z_\alpha$

we reject H_0 if $Z < -Z_\alpha$

(α will be given)

Remark

نفس الاشي لل t-tabel

بس بدل $\alpha \to \frac{\alpha}{2}$, n-1

$Z \to t$

بدل $Z_{\frac{\alpha}{2}} \to t_{\frac{\alpha}{2}}$

Application :

Ex(1) : suppose we want to show that Children have an avarge higher cholesterol Level that the national avarage. It is known that the mean cholesterol Leveel for all Americans is 190.

We test 100 only childern and find that $\bar{x} = 98$

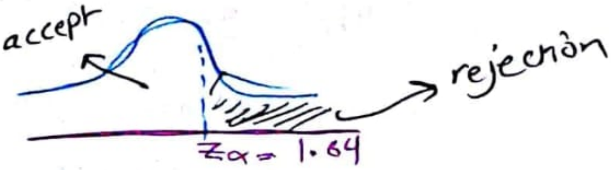
$(\alpha = .05)$ $S = 15$

100 → Z-table اكبر من 30 معناتو مباشرة

Step 1 : $H_0 : \mu = 190$

$H_1 : \mu > 190$

Step 2 :



$Z_\alpha = Z_{0.05} = 1.64$

critical point

step (3)

since $n = 100 > 30$
large

$$Z_{statistic} = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

$$= \frac{198 - 190}{15/\sqrt{100}}$$

$$= 5.33$$

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we reject H_0

الاستنتاج :

We can conclude that the childern have a higher average choletrol level than the national average.

ch 7

* مهمة بالفاينال

$90\%\ C.I \Rightarrow Z_{\frac{\alpha}{2}} = \boxed{Z_{.05}} = 1.64$

$95\%\ C.I \Rightarrow Z_{\frac{\alpha}{2}} = \boxed{Z_{.025}} = 1.96$

$99\%\ C.I \Rightarrow Z_{\frac{\alpha}{2}} = \boxed{Z_{0.005}} = 2.57$

significance level $= \alpha$

ص 21

Ex: $\mu_0 = 4.2$

$n = 25$

$\bar{X} = 4.7$

$S = 1.1$

$\alpha = .05$

الحل

① Teste $H_0 : \mu \leq 4.2 \rightarrow \mu_0$ باستخدام الدواء القديم

معدل البقاء للمرضى باستخدام الدواء الجديد

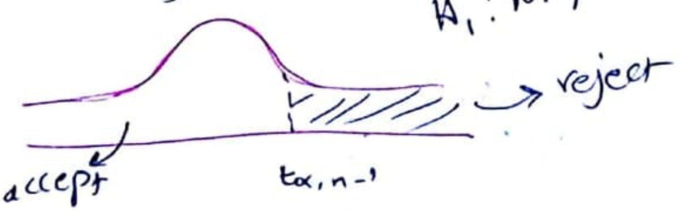
$H_1 : \mu > 4.2$

② since $n = 25 < 30$ small

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{4.7 - 4.2}{1.1/\sqrt{25}} = 2.27$$

③ Determin Region of critical Point.
rejection

$H_1 : \mu > 4.2 \rightarrow$ right tailed



$t_{0.05, 24} = 1.711$

الاستنتاج: * since $t = 2.27 > 1.711 \rightarrow$ reject $H_0 \rightarrow$ accept H_1

$\mu > 4.2$

Ex:

$H_0 : \mu = 35$

$H_1 : \mu \neq 35$

$\alpha = .05$

$n = 14$

$\bar{X} = \frac{\sum_{i=1}^{14} x_i}{14} = 30.5$

$S^2 = 10.64$

sd $n = 14 < 30$ → small

$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

$= \frac{30.5 - 35}{10.64/\sqrt{14}}$

$= -1.58$

$-t_{\frac{\alpha}{2}, n-1}$ $\qquad$ $t_{\frac{\alpha}{2}, n-1}$

$-t_{.025, 13}$ $\qquad$ $t_{.025, 13}$

-2.16 $\qquad$ 2.16

since

$t = -1.58$

$2.16 < t < 2.16$

we accept H_0

$\therefore \mu_0 = 35$

Ex

$n=40$

$\bar{x}=51$

$\sigma=2$

histroy

$\mu=50$

① Test : $H_0: \mu = 50$

$H_1: \mu \neq 50$

② $n=40$, $n>30$, large

$$Z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} = \frac{51-50}{2/\sqrt{40}} = 3.16$$

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reject $\quad -Z_{\frac{\alpha}{2}} = -1.96 \qquad Z_{\frac{\alpha}{2}} = Z_{.025} = 1.96$

since $Z = 3.16 > 1.96$

reject H_0

يعني قبول H_1

معناها في تغير

Ex: A sample of $n=16$ provides a sample mean $\bar{x}=11$ and a sample standard deviation $s=3$ Consider the following hypothesis: $H_0: \mu = 10$

$H_1: \mu > 10$

Test the above hypothesis at $\alpha = .05$

Sol

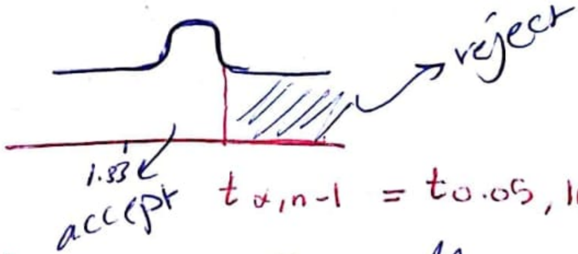
① since $n=16 < 30$ (small)

$$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}} = \frac{11-10}{3/\sqrt{16}} = \frac{1}{3/4} = \frac{4}{3} = 1.33$$

② Determine Rejection Rejon and critical Point

$H_1: \mu > 10$

right tailed



$t_{\alpha, n-1} = t_{0.05, 15} = 1.761$

since $t_{1.33} < 1.761$ we accept H_0

Ex: ص 135 ↑ نفس المعطيات

$n = 100$
$\bar{X} = 1.1$
$S = 0.5$
$H_0 : \mu = 1$
$H_1 : \mu \neq 1$

① $\alpha = 0.05$
② $\alpha = 0.01$

النص الفعلي للإمتحان.

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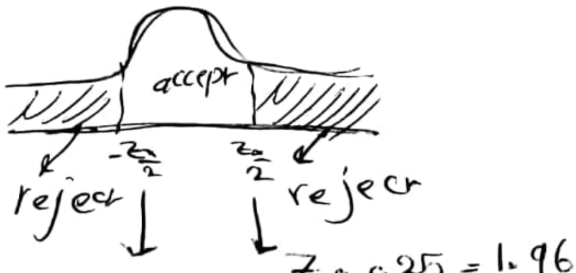
Sol: since $n = 100 > 30$ (large)

$$Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{1.1 - 1}{0.5/\sqrt{100}} = 2$$

$H_1 : \mu \neq 1$
two tailed



-1.96

since $Z = 2 > 1.96$
we reject H_0

② H.W

The P-value Method :-

Def : P-value : is the smallest level of significance (α) which H_0 is rejected.

steps to perform a test of hypothesis using P-value method :-

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Step 1 : state H_0 and H_1 ?

Step 2 : calculate the test statistic (Z-test) when $n \geq 30$ (large)

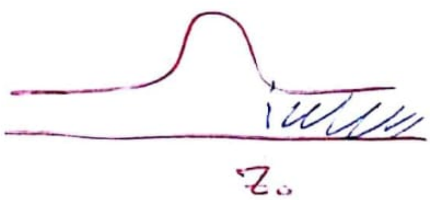
فقط على ال Z-tabel لما تكون n اكبر من 30

$$Z_0 = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

Step 3 : calculate the P-value :-

A) For right tailed

$H_0 : \mu = \mu_0$
$H_1 : \mu > \mu_0$



P-value = Area in right tail

$$= P(Z > Z_0) = 1 - P(Z < Z_0)$$

B) for left tailed :-

$H_0 : \mu = \mu_0$

$H_1 : \mu < \mu_0$

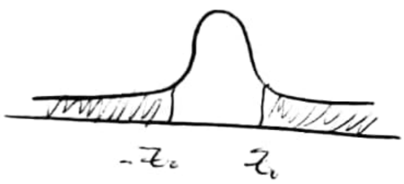
P-value = Area in the left tail

$= P(Z < Z_0)$

c) for two tailed

$H_0 . \mu = \mu_0$

$H_1 : \mu \neq \mu_0$



P-Value = 2 * Area in the Right of Z_0

$= 2 * p(Z > |Z_0|)$

$= 2 * [1 - P(Z < |Z_0|)]$

step 4 : we reject H_0 if P-value $< \alpha$.

we accept H_0 if P-value $\geq \alpha$

Ex: Page (40)

A random sample of size $n = 100$ yields $\bar{x} = 1.1$
$s = 0.5$

Need to test $H_0 : \mu = 1$
$H_1 : \mu \neq 1$ using p-value Method?

Take $\alpha = 0.1$?

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sol

$$Z_0 = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{1.1 - 1}{0.5/\sqrt{100}} = 2$$

$$\text{P-valued} = 2 * P(Z > 2)$$
$$= 2[1 - P(Z < 2)]$$
$$= .0456$$

اذا طلب p- value
بالمستقبل بس بوقف لهون ←

since p-valued $= 0.0456 < 0.1 = \alpha$
we reject H_0

Ex: بصفحة

T بالدفتر 64

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$\alpha = 0.05$

$\mu_0 = 50$

$n = 40 \longrightarrow \bar{X} = 51$ & $S = 2$

need to test $H_0 : \mu = 50$

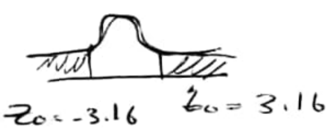
$H_1 : \mu \neq 50$

using P-valued Method ?

Sol :

$$Z_0 = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{51 - 50}{2/\sqrt{40}} = 3.16$$

$$\text{P-valued} = 2 * P(Z > 3.16)$$



$$= 2 * [1 - P(Z < 3.16)]$$

$$= .002$$

since P-value $= .002 < .05 = \alpha$

we reject H_0

$\downarrow$ معناه

we accept H_1

Hypothesis Testing about difference Means $(\mu_1 - \mu_2)$

Method : The rejection region Using $\underline{\underline{t}}$ - statistic

Population I Indepent Population II

mean $= \mu_1 \longrightarrow$ unknown $\longleftarrow$ mean $= \mu_2$

varanice $= \sigma_1^2$ varaince $= \sigma_2^2$

Random sample
$x_1, x_2 --- x_n$
$X_1 : N(\mu_1, \sigma_1^2)$

Random sample $y_1, y_2 --- y_n$
$Y_1 : N(\mu_2, \sigma_2^2)$

$\bar{X} = \dfrac{\sum_{i=1}^{n} x_i}{n}$ $\quad$ $\bar{Y} = \dfrac{\sum_{i=1}^{n} y_i}{n}$

sample mean

[$\bar{X} - \bar{Y}$ is an estimator $\mu_1 - \mu_2$]

Test , $H_0 : \mu_1 = \mu_0$ against $H_1 : \mu_1 \neq \mu_2 \Rightarrow$ Two talied

$\mu_1 > \mu_2 \Rightarrow$ Right

$\mu_1 < \mu_2 \Rightarrow$ left

steps to perform a test of hypothesis

1] using Test - static :-

Assuming equal variance $\sigma_1^2 = \sigma_2^2 = \sigma^2$

state the null hypothesis H_0

& the alternative hypothesis H_1

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2] $n_1 < 30$ & $n_2 < 30$ (n_1 & n_2 small)

compute $t = \dfrac{\bar{x} - \bar{y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$

Where $S_p^2 = \dfrac{(n_1 - 1) S_1^2 + (n_2 - 1) S_2^2}{n_1 + n_2 - 2}$

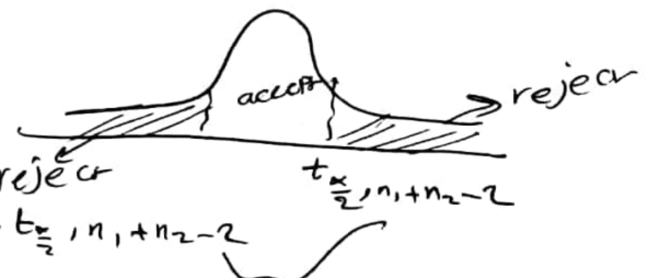
$$S_1^2 = \sum_{i=1}^{n_1} \frac{(x_i - \bar{x})^2}{n_1 - 1} \qquad S_2^2 = \frac{\sum_{i=1}^{n_2} (y_i - \bar{y})^2}{n_2 - 1}$$

3] Determine the Rejection & acceptance Regions & $t_{critical}$:-

A] Two - tailed

$H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 \neq \mu_2$



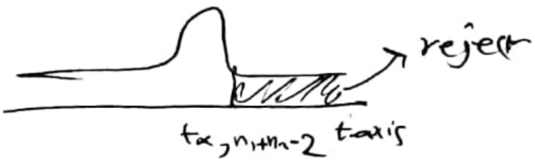
we reject H_0 if $t > t_{\frac{\alpha}{2}, n_1+n_2-2}$

or $t < -t_{\frac{\alpha}{2}, n_1+n_2-2}$

(B) Right tailed

$H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 > \mu_2$

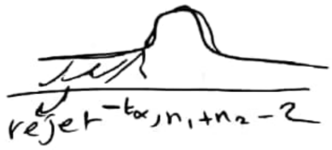


we reject H_0 if $t > t_{\alpha, n_1+n_2-2}$

(C) left taild

$H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 < \mu_2$



we reject H_0 if $t < -t_{\alpha, n_1+n_2-2}$

Ex: suppose Indepedent random samples of 10 English teacher's & 18 Mathematics teachers.

Provided the following statistic about normally :-

English μ_1	Math μ_2
$\bar{X}_1 = 12.93$	$\bar{X}_2 = 13.42$
$S_1 = 2.25$	$S_2 = 2.36$

use $\alpha = 0.05$

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can we conclude that an avarage the Math teacher's more than the English teacher's ?

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sol

1] $H_0 : \mu_1 = \mu_2$

$H_1 :$ ~~$\mu_2 \neq \mu_2$~~ ~~$\mu_2 \neq \mu_1$~~

$\mu_1 < \mu_2$

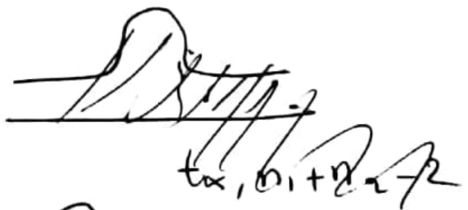
2] $$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{12.93 - 13.42}{2.32\sqrt{\frac{1}{10}+\frac{1}{18}}} = -0.535$$

$$s_p^2 = \frac{s_1^2(n_1-1) + s_2^2(n_2-1)^2}{n_1+n_2-2}$$

$$= \frac{(2.25)^2(10-1) + (2.36)^2(18-1)}{18+10-2} = 5.394$$

$$sp = \sqrt{5.394} = 2.323$$

3] $t_{.05,26} = 1.706$



$-t_{\alpha, n_1+n_2-2} = 1.706$

we accept , H_0 $\mu_1 = \mu_2$

Ex: The following results are for Independent random samples taken from two normal population:-

Assume equal variances:-

$n_1 = 8$, $\bar{x} = 1.4$, $S_1 = 0.4$

$n_2 = 7$, $\bar{Y} = 1.0$, $S_2 = 0.6$

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Test the following hypothesis at $\alpha = 0.05$?

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 > \mu_2$?

sol

comput $t = \dfrac{\bar{x} - \bar{Y}}{sp\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \dfrac{0.4}{.5022(\sqrt{\frac{1}{8} + \frac{1}{7}})} = \dfrac{0.4}{(0.5)\sqrt{0.2678}} = \boxed{1.54}$

where: $s_p^2 = \dfrac{s_1^2(n_1-1) + s_2^2(n_2-1)}{n_1 + n_2 - 2}$

$sp^2 = \dfrac{(0.4)^2(7) + (0.6)^2(6)}{13}$

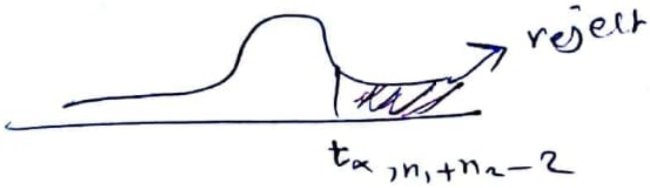
$= 0.2523$

$sp = \sqrt{.2523} = .5022$

8°

$H_1 : \mu_1 > \mu_2$

right tailed



$t_{0.05, 13} = 1.771$

we do not reject H_0

(accept H_0)

$t = 1.54 < 1.771$

سؤال مهم *

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Ex: 52

$H_0 : \mu_1 = \mu_2$

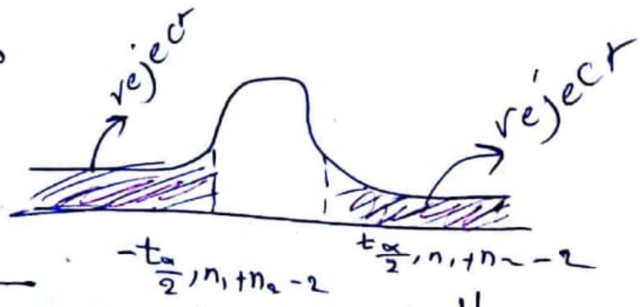
$H_1 : \mu_1 \neq \mu_2$

$\bar{X} = 53.07$, $\bar{Y} = 54.25$, $n_1 = 7$, $n_2 = 6$

$S_1 = 1.91$, $S_2 = 1.71$

sol/ $t = \dfrac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = -1.16$

من القانون ؟؟



$\Downarrow$ -2.201

$\Downarrow$ $t_{0.025, 11} = 2.201$

we accept H_0

$\mu_1 = \mu_2$

Ex ⟶

$H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 < \mu_2$

$\bar{x} = 120$

$\bar{x} = 125$

$n_1 = 6$

$n_2 = 10$

$s_1 = 10$

$s_2 = 9$

نفس الحل

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(III) Paired sample t-test..

* All before and after experiments are paired designs.

Ex:. Testing wither a diet program is effective for weight loss.

Weight of the person befor the program x

" " " " After " " y

هنا فقط المعطى بالأعلى

Subject No	(Befor) weight (x)	(after) weight (y)	Difference
1	x_1	y_1	$d_1 = x_1 - y_1$
2	x_2	y_2	$d_2 = x_2 - y_1$
⋮			
n	x_n	y_n	$d_n = x_n - y_n$

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This gives paired data (dependent sample)

The usual analysis is based on the difference between x & y values $d_i = x_i - y_i$, $i = 1 \dots n$

Fact:.

1. The difference $d_1, d_2, \dots, d_n$ represent a random sample

$$\bar{d} = \frac{\sum_{i=1}^{n} d_i}{n}$$

$$S_d^2 = \frac{\sum_{i=1}^{n} (d_i - \bar{d})^2}{n-1} = \frac{\sum_{i=1}^{n} d_i^2 - n(\bar{d})^2}{n-1}$$

2. Let $d_1, d_2, \dots, d_n$ be a random sample from $N(\mu_d, \sigma_d^2)$

variance of the paired difference of the population.

mean of the paired difference for the population $\mu_d = \mu_1 - \mu_2$

Then the sampling distribution of $\bar{d}$ is:

$\mu_{\bar{d}} = \mu_d$ & $\sigma_{\bar{d}}^2 = \frac{\sigma_d^2}{n}$ $\boxed{\bar{d} : N(\mu_{\bar{d}}, \frac{\sigma_d^2}{n})}$

Test:

$H_0 : \mu_d = 0$
$\mu_1 - \mu_2 = 0$
$(\mu_1 = \mu_2)$

against $H_1 : \mu_d \neq 0 \xRightarrow{\text{Two tailed}} (\mu_1 \neq \mu_2)$

$\mu_d > 0 \xRightarrow{\text{right-tailed}} (\mu_1 > \mu_2)$

$\mu_d < 0 \xRightarrow{\text{left-tailed}} (\mu_2 < \mu_2)$

steps to perform a test of hypothesis:

step(1):
state H_0 & H_1

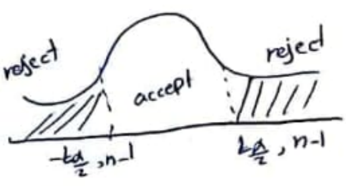
step(2):
Calculate the test of statistic

$$t = \frac{\bar{d} - \overset{\text{zero}}{\mu_d}}{S_d / \sqrt{n}} = \frac{\bar{d}}{S_d / \sqrt{n}}$$

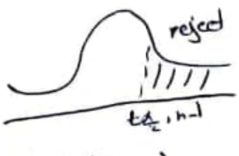
where: $\bar{d} = \frac{\sum_{i=1}^{n} d_i}{n}$

$$S_d = \sqrt{\frac{\sum d_i^2 - n(\bar{d}_i)^2}{n-1}}$$

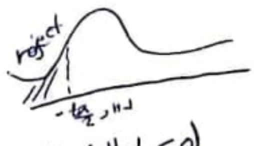
step (3)
Calculate the Critical value and Rejection Rejons:



fig($\mu_d \neq 0$)



fig($\mu_d > 0$)



fig($\mu_d < 0$)

اتخاذ القرار
step(4) Decision ⇒

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Ex: (Page 68)   سؤال للفاينل على 8 علامات.

The sleep hours of 5-patients and after taking ~~a~~ a medication are given by the following:

Subject	Befor	After	d ($x-y$)	d^2
1	6	9	-3	9
2	5	4	1	1
3	7	9	-2	4
4	4	7	-3	9
5	5	6	-1	1
			-8	24

Can you conclude that the medication is effective in increasing the sleep hours? Use $\alpha = 0.01$?

Solution:

step(1)

state the hypothesis

$H_0: \mu_d = 0$

$H_1: \mu_1 < \mu_2$

step (2)

Calcualat t

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} = \frac{-1.6}{1.67/\sqrt{5}} = -2.14$$

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$$\bar{d} = \frac{\sum_{i=1}^{n} d_i}{n} = -\frac{8}{5} = -1.6$$

$$s_d = \sqrt{\frac{\sum d_i^2 - n(\bar{d})^2}{n-1}} = \sqrt{\frac{24 - 5(-1.6)^2}{4}} = 1.67$$

step(3): Calculate the Critical values

$-t_{0.01,4} = -3.477$

step(4): Since $t = -2.14 > -3.477 \Rightarrow$ we accept H_0

$\mu_d = 0 \Rightarrow \mu_1 = \mu_2$

No evidence that the Medication is effective increasing the sleep.

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Chapter 8: Simple linear Regression and correlation:

Defn:

1. Simple linear regression: is a method used to describe the relationship between two variables.
2. regression line: the best fit line
3. Scatter plot = A plot of Data values on a Coordinates System

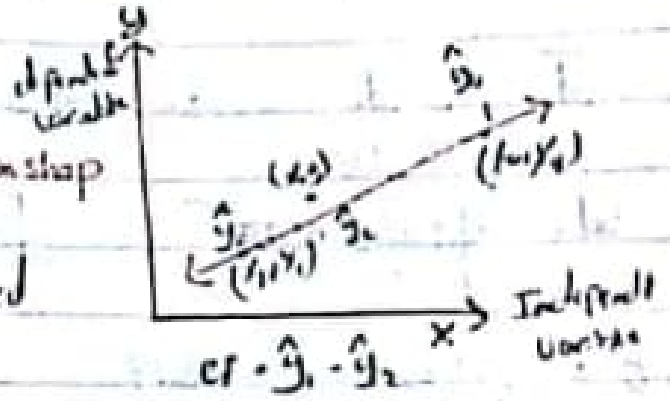
Ex x: hours studies

Y: test Score

Assuming linear relationship

$\hat{y} = a + bx$

where $\hat{y}$ - the fitted (predicted values)



$$b = \frac{S_{xy}}{S_{xx}}, \quad a = \bar{y} - b\bar{x}$$

$$S_{xy} = \sum xy - \frac{\sum x \cdot \sum y}{n}$$

Sum of square

$$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}$$

$$S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n}$$

x	y	xy	x²	y²

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Ex Given the Following Data

x	5	1	7	9	2	2
y	16	15	4	23	14	21

Given $\bar{x} = 6$, $\bar{y} = 18$, $S_{xx} = 88$, $S_{yy} = 64$, $S_{xy} = 67$.
Find the Regression line?

soln: $\hat{y} = a + bx$

$b = \frac{S_{xy}}{S_{xx}} = \frac{67}{88} = 0.761$

$a = \bar{y} - b\bar{x}$
$= 18 - 0.761 * 6$
$= 13.432$

$\boxed{\hat{y} = 13.432 + 0.761 X}$

$\hat{y} = 13.432 + 0.761 (5)$
$= 17.23$

16 من جدول

\# Pearson Correlation Coefficient (r):
Is a measure the strenght bet of the linear relationship between two varible for a sample and given be :-

$r = \frac{S_{xy}}{\sqrt{(S_{xx}) \cdot (S_{yy})}}$, $-1 \le r \le 1$

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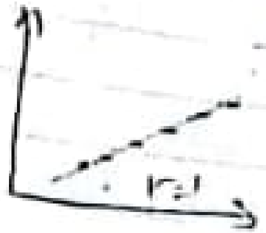
Properties of Person Correlation (r) :-

1. If $r=0$ → there is no linear relationship between the two variable

$r=0$

2. If $r=1$ or $r=-1$?

All Points lies exactly on a straight line.



③ Small low (Correlation) $0.1 < |r| \le 0.3$

Medium " : $0.3 < |r| \le 0.5$

Large (High) " : $0.5 < |r| \le 1$

<u>Ex</u> In Previous Example:

Compute the Pearson's Correlation coefficient (r):

$$r = \frac{S_{xy}}{\sqrt{(S_{xx})(S_{yy})}} = \frac{67}{\sqrt{55 \times 64}} = \frac{67}{\sqrt{5632}} \approx .9$$

(Large

Jordan University of Science & Technology
Department of Mathematics & Statistics
Math 131 (Introduction to Statistics)
SIMPLE LINEAR REGRESSION
Lecture 14

Instructor: Mr. Issam Abu-Irwaq

SIMPLE LINEAR REGRESSION

REGRESSION EQUATION(LEAST SQUARE REGRESSION LINE)

The regression equation is given by $\hat{y} = a + bx$ where

$\boxed{b = \frac{S_{xy}}{S_{xx}}}$, $\boxed{a = \bar{y} - b\bar{x}}$ where

$\boxed{S_{xy} = \sum xy - \frac{(\sum x)(\sum y)}{n}}$ and $\boxed{S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}}$

and $\boxed{S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n}}$, S stands for sum of squares.

LINEAR CORRELATION

VALUE OF THE CORRELATION COEFFICIENT

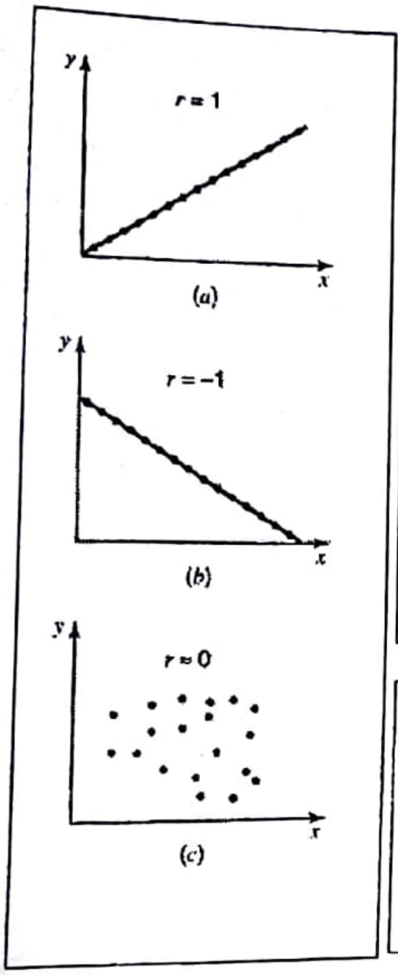
The value of the correlation coefficient always lies in the range -1 to 1; that is,

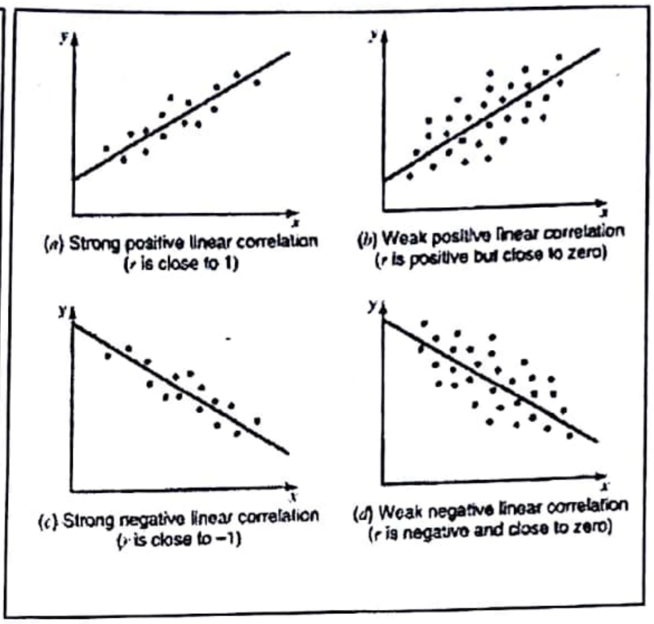
$$-1 \leq r \leq 1$$

LINEAR CORRELATION COEFFICIENT

The simple linear correlation, denoted by r, measures the strength of the linear relationship between two variables for a sample and is calculated as

$$\boxed{r = \frac{S_{xy}}{\sqrt{S_{xx} \times S_{yy}}}}$$





Linear correlation between two variables

Example 1: The following data are the number of hours which 5 persons studied for a statistics and their scores on the test:

Hours Studies (x)	4	9	10	12	4
Test score (y)	30	60	65	70	40

a) Find the regression equation

b) Predict the average test score of a person who studies 14 hours for the test

c) Compute the correlation coefficient (r)

Solution:

The following steps are performed to compute a and b

Step 1: create the following table

x	y	x^2	y^2	xy
4	30	16	900	120
9	60	81	3600	540
10	65	100	4225	650
12	70	144	4900	840
4	40	16	1600	160
$\sum x = 39$	$\sum y = 265$	$\sum x^2 = 357$	$\sum y^2 = 15225$	$\sum xy = 2310$

Step 2: Compute $\overline{x}$ and $\overline{y}$

$$\overline{x} = \frac{\sum x}{n} = \frac{39}{5} = 7.8$$
$$\overline{y} = \frac{\sum y}{n} = \frac{265}{5} = 53$$

Step 3: Compute S_{xy} , S_{xx}, S_{yy}

$$S_{xy} = \sum xy - \frac{\left(\sum x\right)\left(\sum y\right)}{n} = 2310 - \frac{(39)(265)}{5} = 243$$

$$S_{xx} = \sum x^2 - \frac{\left(\sum x\right)^2}{n} = 357 - \frac{(39)^2}{5} = 52.8$$

$$S_{yy} = \sum y^2 - \frac{\left(\sum y\right)^2}{n} = 15225 - \frac{(265)^2}{5} = 1180$$

Step 4: Compute a and b

$$b = \frac{S_{xy}}{S_{xx}} = \frac{243}{52.8} = 4.6$$
$$a = \overline{y} - b\overline{x} = 53 - (4.6)(7.8) = 17.12$$

a) Find the regression equation (Least square Regression line)

Thus, our estimated regression model $\hat{y} = 17.12 + 4.6x$

b) Predict the average test score of a person who studies 14 hours for the test

$$\hat{y}\Big|_{x=14} = 17.12 + 4.6(14) = 81.52$$

c) **Compute the correlation coefficient (r)**

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{243}{\sqrt{(52.8)(1180)}} = 0.97$$

Example 2: The Data in the following table represent final scores in two courses for a sample of seven students in Accounting and Business statistics

Accounting (x)	Business Statistics (y)
73	82
95	87
57	69
43	37
25	32
94	100
66	57

$n=7$

a) Find the regression line $\hat{y} = a + bx$ with Accounting as an independent variable and Business Statistics as a dependent variable. =

$a = 5.82$
$b = .93$
$\hat{y} = 5.82 + 0.93x$

b) Using the regression equation, make estimate Business statistics score if Accounting score is 86? = $x = 80$

c) Compute the correlation coefficient (r) $r =$

Example 3: We wish to explore the relationship between family monthly food consumption (y) and family monthly income (x), both measured in hundreds of dollars, we are given information for 100 families in the following table

Family food consumption versus monthly income		
$\bar{x} = \$8$	$\bar{y} = \$6$	$\sum x^2 = 10000$
$\sum y^2 = 4500$	$\sum xy = 6000$	$n = 100$

a) Find the estimating equation $\hat{y} = a + bx$? $\hat{y} = 3.4 + 0.33x$

b) Using the regression equation, make estimate consumption for a family with a monthly income of $1500? $x = 1500$

c) Compute the correlation coefficient (r)

$r =$

H.W

1) Assume the population are normally distribution, Make the following test of hypothesis :-

(a) $H_0: \mu = 80$
$H_1: \mu \neq 80$, $n = 33$, $\bar{x} = 76.5$, $\sigma = 15$
α, 0.1

(b) $H_0: \mu = 32$
$H_1: \mu < 32$, $n = 75$, $\bar{x} = 26.5$, $S = 7.4$
$\alpha = 0.01$

(c) $H_0: \mu = 55$
$H_1: \mu > 55$, $n = 40$, $\bar{x} = 60.5$, $S = 9$
$\alpha = 0.05$

(d) $H_0: \mu = 60$
$H_1: \mu \neq 60$, $n = 14$, $\bar{x} = 57$, $S = 9$
$\alpha = 0.05$

(e) $H_0: \mu = 35$
$H_1: \mu < 35$, $n = 24$, $\bar{x} = 28$
$S = 5.4$, $\alpha = 0.005$

(f) $H_0: \mu = 47$
$H_1: \mu > 47$, $n = 18$, $\bar{x} = 50$
$S = 6$, $\alpha = .001$?